

Monte-Carlo calculation of conversion factors for the estimation of mean glandular breast dose

To cite this article: D R Dance 1990 *Phys. Med. Biol.* **35** 1211

View the [article online](#) for updates and enhancements.

You may also like

- [Normalized mean glandular dose computation from mammography using GATE: a validation study](#)
Marios E Myronakis, Marketa Zvelebil and Dimitra G Darambara
- [Computing mammographic density from a multiple regression model constructed with image-acquisition parameters from a full-field digital mammographic unit](#)
Lee-Jane W Lu, Thomas K Nishino, Tuenchit Khamapirad et al.
- [Estimation of mean glandular dose for breast tomosynthesis: factors for use with the UK, European and IAEA breast dosimetry protocols](#)
D R Dance, K C Young and R E van Engen



physicsworld WEBINAR

ZAP-X radiosurgery & ZAP-Axon SRS planning

Technology Overview, Workflow, and Complex Case Insights from a Leading SRS Center

Get an inside look at European Radiosurgery Center Munich – a high-volume ZAP-X centre – with insights into its vault-free treatment suite, clinical workflow, patient volumes, and treated indications. The webinar will cover the fundamentals of the ZAP-X delivery system and what sets it apart from other SRS platforms; showcase real-world performance through complex clinical cases; and provide a concise overview of the recently unveiled next-generation ZAP-Axon radiosurgery planning system.

LIVE at 4 p.m. GMT/8 a.m. PST, 19 Feb 2026

[Click to register](#)

Monte Carlo calculation of conversion factors for the estimation of mean glandular breast dose

D R Dance

Department of Physics, Royal Marsden Hospital, Fulham Road, London SW3 6JJ, UK

Received 26 March 1990, in final form 9 May 1990

Abstract. The IPSM report on the commissioning and routine testing of mammographic x-ray systems recommends that breast dose be specified as the mean dose to the glandular tissues within the breast and gives the size and compositions of a standard breast phantom for the comparison of doses. The dose to this standard breast phantom can be determined by measuring the incident air kerma to a Perspex phantom and applying appropriate multiplicative conversion factors. These conversion factors have been evaluated by Monte Carlo calculations for a wide range of mammographic x-ray spectra. Some factors are provided for a range of breast thicknesses to supplement existing tabulations. Results are also given for equivalent thicknesses of Perspex and breast tissue.

1. Introduction

The female breast is a radiosensitive organ and there is a risk of carcinogenesis associated with the mammographic examination. With modern equipment and technique this risk is small (NCRP 1986, Law 1987), but it is important that it be minimised, and an assessment of the dose to the breast should be included in any mammographic quality assurance programme (IPSM 1989).

The x-ray spectra used for mammography are low energy and the depth dose within the breast decreases rapidly with increasing depth (Hammerstein *et al* 1979). Breast dose is specified, therefore, using a quantity which is representative of the dose to the whole organ. It is believed that the glandular tissues within the breast (including acinar and ductal epithelium and associated stroma) are the most sensitive to radiation-induced carcinogenesis and Hammerstein *et al* (1979) suggested that the mean dose to the glandular tissues within the breast is a suitable dosimetric quantity. This suggestion has been taken up widely: the use of mean glandular dose has been recommended by the NCRP (1986) and the ICRP (1987). It has also been adopted in the recently published IPSM report on quality assurance in mammography (IPSM 1989).

It is difficult to measure the mean glandular dose to the breast directly and it is usual to employ conversion factors which relate the incident air kerma to this dose. Such factors have been measured by some authors (e.g. Hammerstein *et al* 1979, Stanton *et al* 1984) and have been calculated by others using Monte Carlo techniques (e.g. Dance 1980, Rosenstein *et al* 1985) for the conventional x-ray spectra used in screen/film or xero- mammography. No factors are presently available for the spectra from a tungsten target filtered with K-edge filters which are sometimes used in screen-film mammography (Beaman *et al* 1983, Säbel *et al* 1986). The most comprehensive of the existing tabulation of factors are those of Stanton *et al* (also given in NCRP

85, 1986) and Rosenstein *et al* (1985). The tables of Stanton *et al* have some data missing for compressed breast thicknesses of 3, 7 and 8 cm. They are based on the integration of depth-dose curves and do not account for variation of the dose laterally within the breast. The tables of Rosenstein *et al* provide data for a range of breast shapes but factors for firm compression are only given for a thickness of 6 cm.

Breast dose varies widely with breast composition and thickness as well as the choice of imaging equipment and radiographic technique. To facilitate a comparison of variations due to the imaging system alone, the IPSM (1989) has developed a protocol for estimating the dose to a 4.5 cm thick standard breast from measurements using a Perspex phantom, and appropriate conversion factors. This paper describes the Monte Carlo calculation of these conversion factors for a wide range of x-ray spectra including K-edge filtered spectra from both molybdenum and tungsten targets. For comparison purposes, and in order to supplement the existing tabulations of Stanton *et al* (1984) and Rosenstein *et al* (1985), the incident air kerma to mean glandular dose conversion factor has also been evaluated for breast thicknesses in the range 2–8 cm.

Perspex is a cheap and convenient material for constructing breast phantoms and the Monte Carlo program has also been used to calculate equivalent thicknesses of Perspex and breast tissue which will be of value in the practical assessment of the performance of breast imaging systems for different breast sizes.

2. Dose estimation for a standard breast

The standard breast adopted by the IPSM (1989) is shown in figure 1. It is a cylinder 4.5 cm thick with semi-circular cross section and diameter 16 cm. These dimensions were chosen to be representative of an average-sized breast undergoing firm compression. The area of the breast, however, is not critical as the dose shows quite a small dependence on this quantity (Dance 1980).

The standard breast has a central region which is composed of a 50:50 mixture by weight adipose and glandular tissues and an outer shield region of adipose tissue 0.5 cm thick. In the practical situation it is difficult to construct a phantom according to the above prescription and it is suggested instead that the mean glandular dose be estimated from measurements of the incident air kerma to a 4 cm thick Perspex phantom

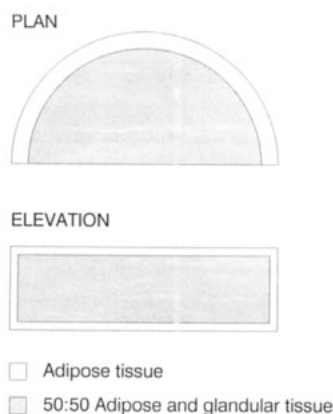


Figure 1. The standard breast phantom, showing the positions of the central and adipose regions.

with the same cross-sectional area. The method involves the determination of the incident air kerma (K) to the Perspex phantom (measured without backscatter and with the compression cone in place) and the half value layer of the x-ray set (also measured with the compression cone in place). Appropriate measurement techniques are described in the IPSM report (1989). The mean glandular dose to the standard breast can then be calculated using the prescription

$$D = Kpg \quad (1)$$

where p converts the incident air kerma to the Perspex phantom to that for the standard breast and g converts the incident air kerma for the standard breast to mean glandular dose. Both of these conversion factors depend upon radiation quality.

To make the work more generally useful, some calculations have also been made for other breast thicknesses. In all cases the tissue compositions and densities have been based on the data given by Hammerstein *et al* (1979).

3. Method

3.1. The model exposure and the Monte Carlo method

The computer program used for the calculation of the conversion factors p and g (equation (1)) was based on Monte Carlo simulation routines developed previously for the estimation of absorbed dose and scatter in mammography (Dance 1980, Dance and Day 1984). An outline of the method is given below with particular attention being paid to some small changes in methodology and imaging geometry.

Figure 2 shows the imaging geometry used. The breast was compressed at a focus-film distance of 60 cm using a 3 mm thick Perspex compression cone. It had an adipose shield region 0.5 cm thick (section 2) and a central region which was a homogeneous mixture of adipose and glandular tissues. The breast rested directly on the image receptor which was chosen to simulate the Kodak Min-R screen and was a layer of $\text{Gd}_2\text{O}_3\text{S}$ 33.9 mg cm^{-2} thick (A Haus, private communication). Backscatter from material beyond the receptor and sidescatter from regions of the patient away from the breast were neglected (it is estimated from previous work that the error in p and g from this approximation is less than 2%). When appropriate, the breast was replaced by a Perspex slab with the same cross-sectional area.

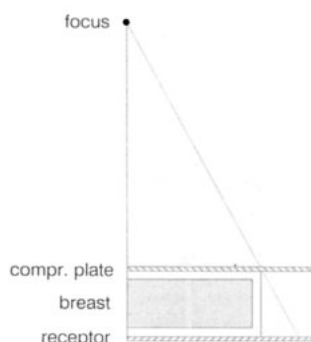


Figure 2. The model used in the Monte Carlo calculations, showing the geometric configuration for simulating the irradiation of the standard breast phantom.

It was assumed that the photons could undergo photoelectric absorption, coherent scatter or incoherent scatter. The total cross section for the former process was taken from the tabulation of Storm and Israel (1970) and the coherent form factors and incoherent scattering functions were taken from Hubbell *et al* (1975) and Hubbell and Øverbø (1979).

Each photon simulated by the Monte Carlo program started at the focal spot of the x-ray tube and its path was traced from region to region of the model until all its energy was absorbed or it left the system. All energy deposited in each region of the model was recorded. In order to reduce computation time, the energy absorption efficiency of the receptor for each incident photon was estimated analytically by interpolation in tables previously calculated (Dance and Day 1984). Sufficient photon histories were generated to achieve a statistical uncertainty of 2% or less in quantities of interest.

As an independent check of the calculations for the image receptor, the ratio of the scattered and primary energies deposited in the receptor was compared with the results of another Monte Carlo program. This program was developed separately for a different purpose (Alm Carlsson *et al* 1989) and used a different computational algorithm (the collision density estimator). The results were in agreement within statistical errors (2%).

3.2. Calculation of the conversion factor g

The mean dose to the glandular tissues was calculated from the energies absorbed in the central region of the breast. The energy deposited following each interaction in this region was shared between the glandular and adipose tissues in proportion to the contribution made by these tissues to the appropriate interaction cross section.

The computation of the incident air kerma was facilitated by the addition of a flat plate ionisation chamber to the model. The chamber, which was 2 mm thick, was placed between the compression plate and the top surface of the breast, and the air kerma was calculated analytically for the first passage of a primary or forward scattered photon through the chamber. The contributions from backscattered photons were ignored. This procedure simulated an air kerma measurement without the patient.

The conversion factor g was estimated as the ratio of the energy absorbed in the glandular tissues to the product of the incident air kerma (without backscatter) and the mass of the glandular tissue present in the central region of the breast.

3.3. Calculation of the conversion factor p

The conversion factor p was estimated from calculations of the energy absorbed in the image receptor per unit incident air kerma (without backscatter). If E_1 and E_2 are the values of this quantity calculated for a 4 cm Perspex slab and for the 4.5 cm thick 50:50 composition standard breast, then p is simply the ratio (E_1/E_2). This relationship is based on the assumption that properly exposed radiographs require the same energy absorption per unit area in the central portion of the image (which corresponds to the most radiologically dense region of the breast). This is a valid assumption for the present case because E_1 and E_2 are similar in magnitude and the effect of reciprocity law failure is small. The energies E_1 and E_2 were calculated for the shadow on the image plane of the bottom of the central region of the breast. This choice excludes penumbral effects due to the diverging beam.

Modern mammographic equipment uses an anti-scatter grid and the effects of removing scatter from the image was investigated by evaluating p using the sum of the primary and secondary energies absorbed in the receptor (no grid case) and just the primary photons (perfect grid case).

3.4. Calculation of equivalent breast and Perspex thicknesses

The calculations of the quantities E_1 and E_2 were also used to estimate the thickness of Perspex equivalent to breast thicknesses in the range 2–8 cm. Thicknesses were taken to be equivalent when the relationship $E_1 = E_2$ was satisfied.

3.5. X-ray spectra

The 90 different x-ray spectra used for the computation of p and g are outlined in table 1. They were chosen to more than encompass the qualities used for screen/film and xero- mammography. Spectra from both tungsten and molybdenum targets were taken from the tabulation of Birch *et al* (1979) and were modified by the addition of molybdenum, rhodium or palladium K-edge filters where appropriate. The half value layers of the spectra were adjusted by filtering with aluminium and spectra for potentials intermediate between tabulated values were obtained by using the interpolation technique described in Dance (1987). For a given target/filter/half value layer combination, the tube potential was typically varied in steps of 1 kV (below 35 kV) and 5 kV (above 35 kV). In some cases, only the values near the ends of a potential range were used, as these determined the range of the conversion factors.

The equivalent Perspex thicknesses have only been calculated for x-ray spectra from a molybdenum target with a 30- μ m molybdenum filter at 28 kV and for half value layers of 0.30, 0.35 and 0.40 mm Al.

Table 1. X-ray spectra used for the calculation of the conversion factors.

HVL range mm Al	Target and filter	kV range
0.25–0.45	Mo+30 μ Mo	25–35
0.45–0.70	W+60 μ Mo	23–35
0.50–0.80	W+50 μ Rh	24–35
0.55–0.90	W+50 μ Pd	25–35
0.50–2.00	W+Al	23–50

4. Results and discussion

The results for the conversion factors g and p and the equivalent thicknesses of Perspex and breast tissue are given in tables 2, 3 and 4 respectively.

The data in table 2 represent the average conversion factors for all the spectra which were used at each half value layer. The spread in the conversion factor at fixed half value layer varied with the thickness of the breast and was typically $\pm 1\%$ for a 2 cm thick breast, $\pm 3\%$ for a 4.5 cm thick breast and $\pm 5\%$ for an 8 cm thick breast. These differences were not felt to be important and the central values only have been tabulated.

Table 2. The conversion factor g which relates incident air kerma (without backscatter) to mean glandular dose for the 'standard' breast phantom.

HVL mm Al	g (mGy mGy ⁻¹) for breast thicknesses of							
	2 cm	3 cm	4 cm	4.5 cm	5 cm	6 cm	7 cm	8 cm
0.25	0.339	0.234	0.174	0.155	0.137	0.112	0.094	0.081
0.30	0.390	0.274	0.207	0.183	0.164	0.135	0.114	0.098
0.35	0.433	0.309	0.235	0.208	0.187	0.154	0.130	0.112
0.40	0.473	0.342	0.261	0.232	0.209	0.172	0.145	0.126
0.45	0.509	0.374	0.289	0.258	0.232	0.192	0.163	0.140
0.50	0.543	0.406	0.318	0.285	0.258	0.214	0.177	0.154
0.55	0.573	0.437	0.346	0.311	0.287	0.236	0.202	0.175
0.60	0.587	0.466	0.374	0.339	0.310	0.261	0.224	0.195
0.65	0.622	0.491	0.399	0.363	0.332	0.282	0.244	0.212
0.70	0.644	0.514	0.421	0.384	0.352	0.300	0.259	0.227
0.80	0.682	0.555	0.460	0.422	0.389	0.333	0.289	0.254
0.90	0.721	0.592	0.500	0.473	0.430	0.378	0.327	0.293
1.00	0.733	0.623	0.534	0.497	0.464	0.407	0.360	0.321
1.20	0.777	0.675	0.588	0.550	0.516	0.456	0.406	0.364
1.40	0.813	0.717	0.632	0.594	0.559	0.497	0.444	0.399
1.60	0.842	0.753	0.670	0.632	0.596	0.533	0.479	0.432
1.80	0.865	0.783	0.704	0.666	0.631	0.567	0.511	0.463
2.00	0.886	0.810	0.734	0.696	0.660	0.596	0.540	0.490

Table 3. The conversion factor p which relates incident air kerma (without backscatter) for the 4 cm Perspex phantom to the incident air kerma (without backscatter) for the 'standard' breast phantom.

HVL mm Al	p	HVL mm Al	p
0.25	1.12	0.70	1.06
0.30	1.10	0.80	1.05
0.35	1.10	0.90	1.04
0.40	1.09	1.00	1.03
0.45	1.09	1.20	1.03
0.50	1.09	1.40	1.03
0.55	1.07	1.60	1.02
0.60	1.06	1.80	1.02
0.65	1.06	2.00	1.02

The results in table 2 have been compared with the most comprehensive of the existing tabulations, namely the measurements of Stanton *et al* (1984) and the calculations of Rosenstein *et al* (1985). Figure 3 shows this comparison for a 6 cm thick breast where data from both of these data sets are available. In total, a comparison could be made with 52 existing data points. Exact agreement for this comparison was not expected because of methodological differences, but generally the comparison was considered to be quite good. Of the 52 data points compared, 36 showed agreement within 5% and only 3 showed a difference of more than 8%. The largest discrepancies (12% and 13%) were for the data of Stanton *et al* at a half value layer of 1.6 mm aluminium and for 7 cm and 8 cm thick breasts. It is believed that this difference arises

Table 4. Equivalent thicknesses of breast tissue and Perspex. The breast tissue has a composition equal parts by weight adipose and glandular tissue and has a 0.5 cm 'adipose shield'.

Breast thickness (cm)	Perspex thickness (cm)	Breast thickness (cm)	Perspex thickness (cm)
2.0	1.74	5.0	4.61
3.0	2.69	6.0	5.54
4.0	3.65	7.0	6.53
4.5	4.15	8.0	7.47

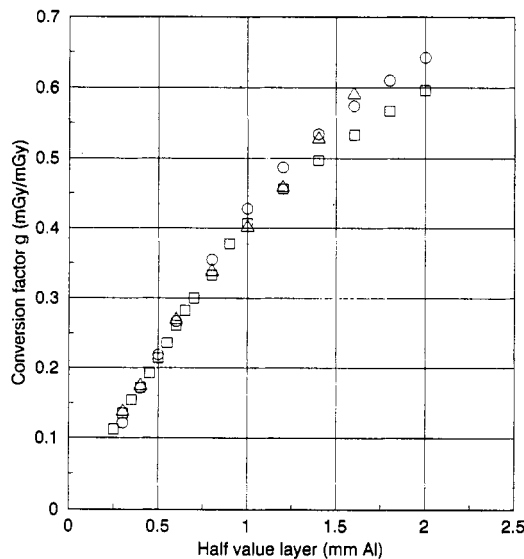


Figure 3. The conversion factor g for a 6 cm thick breast, showing the variation with half value layer. The open squares are the present work, the open circles are the results of Rosenstein *et al* (1985) and the open triangles the results of Stanton *et al* (1984).

because the results of Stanton *et al* are based on the integration of central axis depth doses, no allowance being made for the lateral variation of dose. This approximation is expected to produce a small error at low photon energies, where the mean free path between interactions is small, but a larger error as the energy and mean free path increase. The largest discrepancy with the work of Rosenstein *et al* was 9% and occurred at a half value layer of 0.3 mm aluminium for a 6 cm thick breast. It is thought that this may be explained at least partially by the use of different 'shield' regions surrounding the central area of the breast. The present work used a 0.5 cm layer of adipose tissue whereas Rosenstein *et al* used a 0.4 cm layer of skin. For the softest beam qualities, the thickness and composition of this layer have an important effect on the conversion factor g . At the harder beam qualities, the results of Rosenstein *et al* are 7% higher than for the present work. This difference is due to the use of different focus skin distances (about 3%) and the incorporation of the compression cone within the present model (scatter from the cone increases the exposure and the mean breast dose, and the combined effect decreases the factor g by about 4%).

The air kerma conversion factor p (table 3) shows little variation with x-ray spectrum at fixed HVL and was similar for the calculations without a grid and with a perfect grid. The figures tabulated are the means of all values computed and the range of variation associated with each value is typically $\pm 2\%$.

The equivalent Perspex thicknesses in table 4 are averages of the calculations for the grid and no grid cases and for the three x-ray spectra described previously. It is clear from these results that small errors would arise if Perspex were used purely on the basis of its mass per unit area. The variation of the equivalent thickness with x-ray spectrum and choice of grid was at most $\pm 1.5\%$.

5. Discussion and conclusions

The conversion factors for estimating mean glandular breast dose from measurements of incident air kerma can be expressed solely as a function of half value layer for an extremely wide range of x-ray spectra from both tungsten and molybdenum targets.

The value of the conversion factor g depends upon the apportionment of the energy absorbed in the breast between glandular and adipose tissues and is dependent upon the composition of these tissues. The present calculations (like most others) are based on the data of Hammerstein *et al* (1979), which was derived from just five samples of glandular tissue and eight samples of adipose tissue. They can only be regarded therefore as being indicative of the true mean glandular dose for a particular patient, but they do provide nevertheless a valuable means of comparing the doses associated with the use of different beam qualities and imaging equipments.

The conversion factor p is close to unity because an appropriate thickness was chosen for the Perspex phantom.

The equivalent thicknesses of breast tissue and Perspex are different to the values which would be estimated purely on the basis of density. The tabulated values should be useful in the design of phantoms for the practical assessment of the performance of breast imaging systems.

Acknowledgments

The author is grateful for financial support from the Cancer Research Campaign and Medical Research Council under grant no 841 3976 (387/388). He is indebted to the members of the IPSM mammography working party for sharing their expertise and to Reg Davis, Stephen Evans and Colin Jones at The Royal Marsden Hospital, London and Gudrun Alm Carlson, Calle Carlsson and Jan Persliden at the University Hospital, Linköping for valuable discussions.

Résumé

Calcul par la méthode de Monte Carlo des facteurs de conversion pour l'estimation de la dose moyenne à la glande mammaire.

D'après les recommandations du rapport de l'IPSM sur la recette et les tests de routine des systèmes de mammographie à rayon X, la dose au sein doit être spécifiée comme la dose moyenne délivrée aux tissus de la glande mammaire; ce rapport donne les dimensions et la composition d'un fantôme de sein standard permettant la comparaison des doses. La dose délivrée à ce fantôme de sein standard peut-être déterminée en mesurant le kerma incident dans l'air dans un fantôme de perspex puis en utilisant des facteurs de conversion multiplicatifs appropriés. Ces facteurs de conversion ont été évalués par des calculs utilisant la méthode de Monte Carlo pour une large gamme des spectres de rayons X utilisés en mammographie. Pour

compléter les tabulations existantes, quelques facteurs sont donnés pour une gamme d'épaisseurs de sein. Les auteurs donnent aussi des résultats pour des épaisseurs équivalentes de perspex et de tissu mammaire.

Zusammenfassung

Berechnung von Konversionsfaktoren mit Hilfe der Monte Carlo-Methode zur Bestimmung der mittleren Brustdrüsensdosis.

Der ISPM Report zum Aufbau und Routinetest mammographischer Röntgensysteme empfiehlt, daß Brustdosen als mittlere Dosen des Drüsengewebes in der Brust angegeben werden sollen und gibt Größe und Zusammensetzung eines Standardbrustphantoms für den Dosisvergleich an. Die Dosis für dieses Standardbrustphantom kann bestimmt werden durch Messung der Luftkerma für ein Plexiglasphantom und Anwendung der entsprechenden multiplikativen Konversionsfaktoren. Diese Konversionsfaktoren wurden berechnet mit Hilfe der Monte Carlo-Methode für einen weiten Bereich mammographischer Röntgenspektren. Einige Faktoren werden angegeben für eine Reihe von Brustdicken zur Ergänzung vorhandener Tabellen. Für Äquivalentdicken von Plexiglas und Brustgewebe werden ebenfalls Ergebnisse angegeben.

References

- Alm Carlsson G, Dance D R and Persliden J 1989 Grids in mammography: optimisation of the information content relative to radiation risk *University of Linköping Report ULI-RAD-R-059* (Institutionen för radiologi, Linköping University, S-581 85 Linköping, Sweden)
- Beam S, Lillicrap S C and Price J L 1983 Tungsten anode tubes with K-edge filters for mammography *Br. J. Radiol.* **56** 721-7
- Birch R, Marshall M and Ardran G M 1979 *Catalogue of Spectral Data for Diagnostic X-Rays* SRS 30 (London: Hospital Physicists' Association)
- Dance D R 1987 A method for the reconstruction of diagnostic x-ray spectra from measurements of attenuation *Phys. Med. Biol.* **32** 1631-8
- 1980 The Monte Carlo calculation of integral radiation dose in xeromammography *Phys. Med. Biol.* **25** 25-37
- Dance D R and Day G J 1984 The computation of scatter in mammography by Monte Carlo methods *Phys. Med. Biol.* **29** 237-47
- Hammerstein G R, Miller D W, White D R, Masterson M E, Woodard H Q and Laughlin J S 1979 Absorbed radiation dose in mammography *Radiology* **130** 485-91
- Hubbell J H and Øverbø I 1979 Relativistic atomic form factors and photon coherent scattering cross sections *J. Phys. Chem. Ref. Data* **8** 69-105
- Hubbell J H, Veigele W J, Briggs E A, Brown R T, Cromer D T and Howerton R J 1975 Atomic form factors, incoherent scattering functions and photon scattering cross sections *J. Phys. Chem. Ref. Data* **4** 471-538; 1977 *Errata* **6** 615-6
- Institute of Physical Sciences in Medicine 1989 *The Commissioning and Routine Testing of Mammographic X-Ray Systems* Topic group report 59 (York: IPSM)
- International Commission on Radiological Protection 1987 Statement from the 1987 Como meeting of the ICRP, ICRP Publication 52 *Annals ICRP* **17**(4) (Oxford: Pergamon)
- Law J 1987 Cancers induced and cancers detected in a mammography screening programme *Br. J. Radiol.* **60** 231-4
- National Council on Radiation Protection and Measurements 1986 *Mammography: a User's Guide* Report 85 (Bethesda, MD: NCRP)
- Rosenstein M, Andersen L W and Warner G G 1985 *Handbook of Glandular Tissue Doses in Mammography* FDA 85-8239 (Rockville, MD: US Department of Health and Human Services)
- Säbel M, Willgeroth F, Aichinger H and Dierker J 1986 X-ray spectra and image quality in mammography *Electromedica* **54** 158-65
- Stanton L, Villafana T, Day J L and Lightfoot D A 1984 Dosage evaluation in mammography *Radiology* **150** 577-84
- Storm E and Israel H I 1970 Photon cross sections from 1 keV to 100 MeV for elements $Z = 1$ to $Z = 100$ *Nucl. Data Tables A7* 565-681